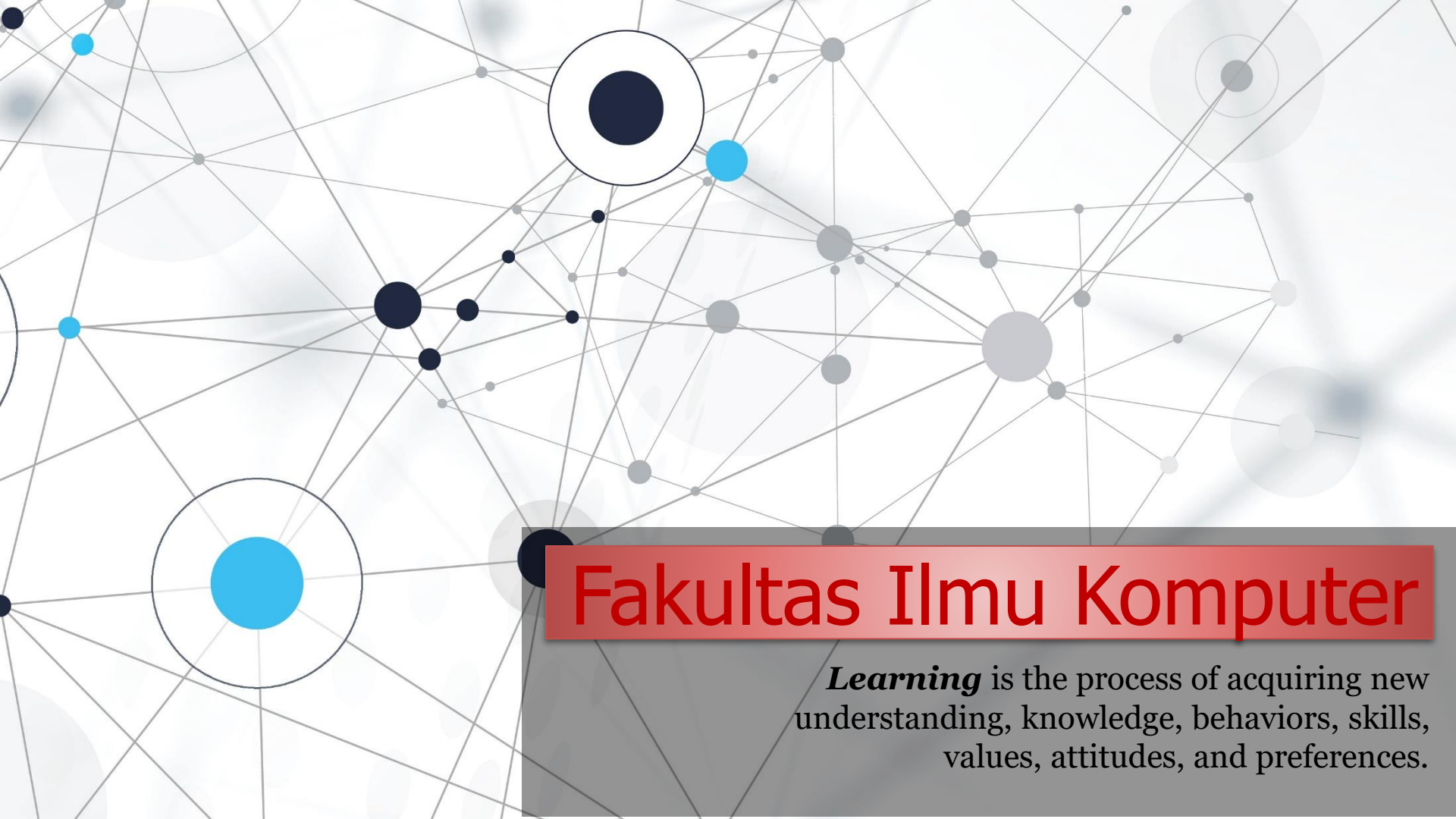


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Back-end Framework #7

[Create Database & Tables]



Welcome to the Back-end Web Development *course*

Three Important MVC Components (*layers*)

Component #1 → Model :

- ✓ The model component stores data and its related logic. It represents data that is being transferred between controller components or any other related business logic. For example, a Controller object will retrieve the customer info from the database. It manipulates data and send back to the database or use it to render the same data.
- ✓ It responds to the request from the views and also responds to instructions from the controller to update itself. It is also the lowest level of the pattern which is responsible for maintaining data.

Component #2 → View :

- ✓ The View is that part of the application that represents the presentation of data.
- ✓ Views are created by the data collected from the model data. A view requests the model to give information so that it resents the output presentation to the user.
- ✓ The view also represents the data from chats, diagrams, and table. For example, any customer view will include all the UI components like text boxes, drop downs, etc.

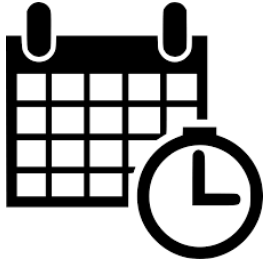
Component #3 → Controller :

- ✓ The Controller is that part of the application that handles the user interaction. The controller interprets the mouse and keyboard inputs from the user, informing model and the view to change as appropriate.
- ✓ A Controller send's commands to the model to update its state (*E.g., Saving a specific document*). The controller also sends commands to its associated view to change the view's presentation (*For example scrolling a particular document*).

The paper was published under the title:

Burbeck, Steve. "Applications programming in smalltalk-80 (*tm*): How to use model-view-controller (mvc)." *Smalltalk-80 v2 5* (1992): 1-11.

Three Important MVC Components (*layers*)



Model: It includes all the data and its related logic.



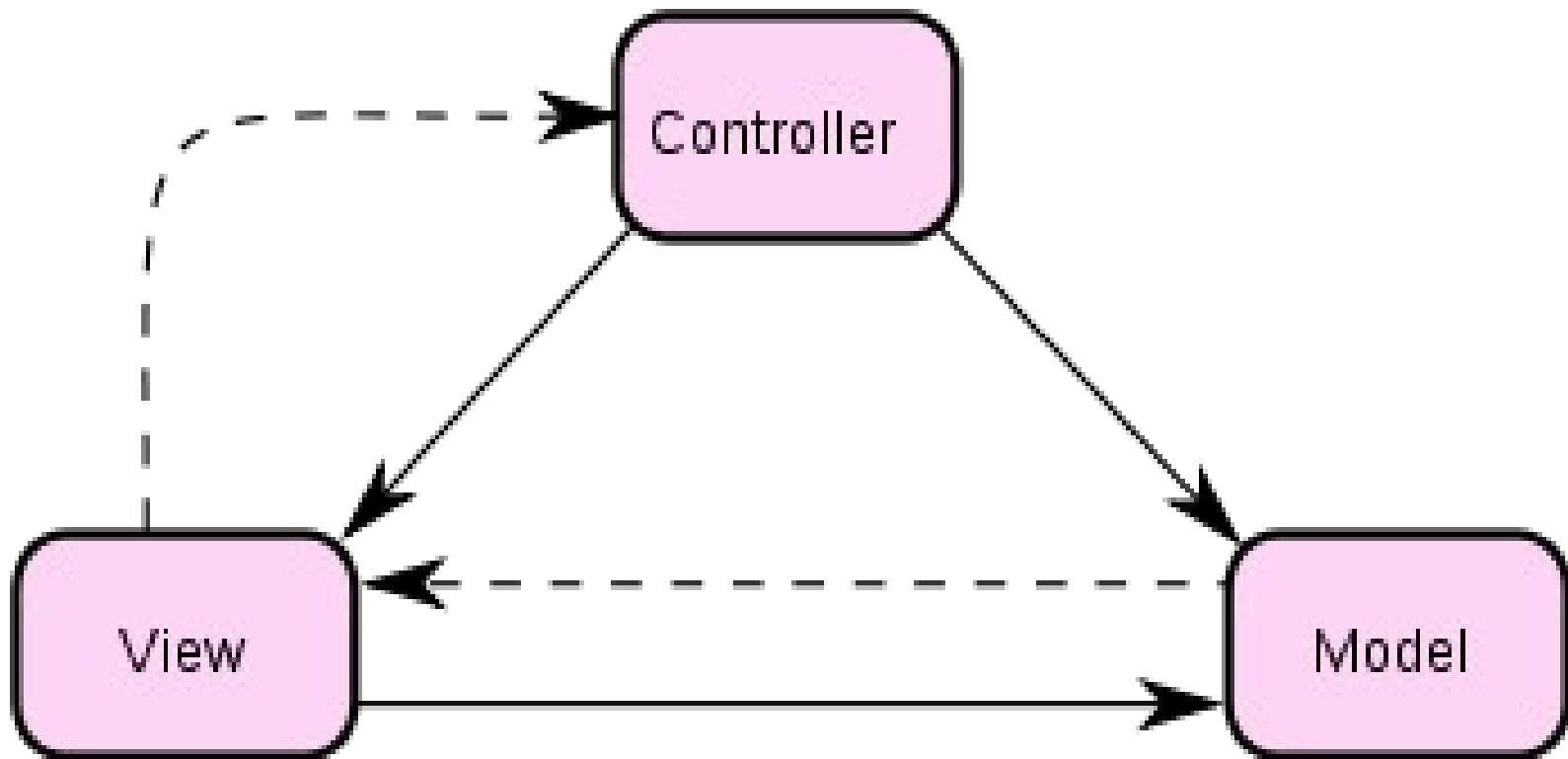
View: Present data to the user or handles user interaction.



Controller: An interface between Model and View components.

Three Important MVC Components (*layers*)

logic program (*controller*)



Tampilan (*view*) => UI

Manage the data

phpMyAdmin in XAMPP

Bringing MySQL to the web

phpMyAdmin is a free software tool written in PHP, intended to handle the administration of MySQL over the Web. phpMyAdmin supports a wide range of operations on MySQL and MariaDB. Frequently used operations (***managing databases, tables, columns, relations, indexes, users, permissions, etc***) can be performed via the user interface, while you still have the ability to directly execute any SQL statement.



Read Source:

<https://www.phpmyadmin.net/>

What is LAMP Stack?



LAMP stack is a popular open-source web development platform. LAMP Stack development is popular web development methodology that allow web developers to build projects quickly, efficiently, and cost-effectively.

The LAMP Stack is a group of open-source software technologies that work together to run dynamic websites and applications. **LAMP**, which stands for **Linux**, **Apache**, **MySQL**, and **PHP**, allows to build websites quickly and easily without buying expensive third-party software licenses.

LAMP Stack Alternatives:

- ✓ **MEAN** stack (MongoDB, Express, Angular, Node. js)
- ✓ **LEMP** stack (Linux, NGINX, MySQL/MariaDB, PHP/Perl/Python)
- ✓ **LAPP** stack (Linux, Apache, PostgreSQL, PHP)
- ✓ **LEAP** stack (Linux, Eucalyptus, AppScale, Python)
- ✓ **LLMP** stack (Linux, Lighttpd, MySQL/MariaDB, PHP/Perl/Python)
- ✓ **XAMPP** stack (Cross-platform, Apache, MariaDB, PHP, Perl)

MySQL

MySQL is a relational database management system based on **SQL – Structured Query Language**. The application is used for a wide range of purposes, including data warehousing, e-commerce, and logging applications. The most common use for mySQL however, is for the purpose of a web database. It can be used to store anything from a single record of information to an entire inventory of available products for an online store. In association with a scripting language such as **PHP** or **Perl**, it is possible to create websites which will interact in real-time with a mySQL database to rapidly display categorised and searchable information to a website user.



Read Source:

<https://www.123-reg.co.uk/support/servers/what-is-mysql-and-why-do-i-need-it/>

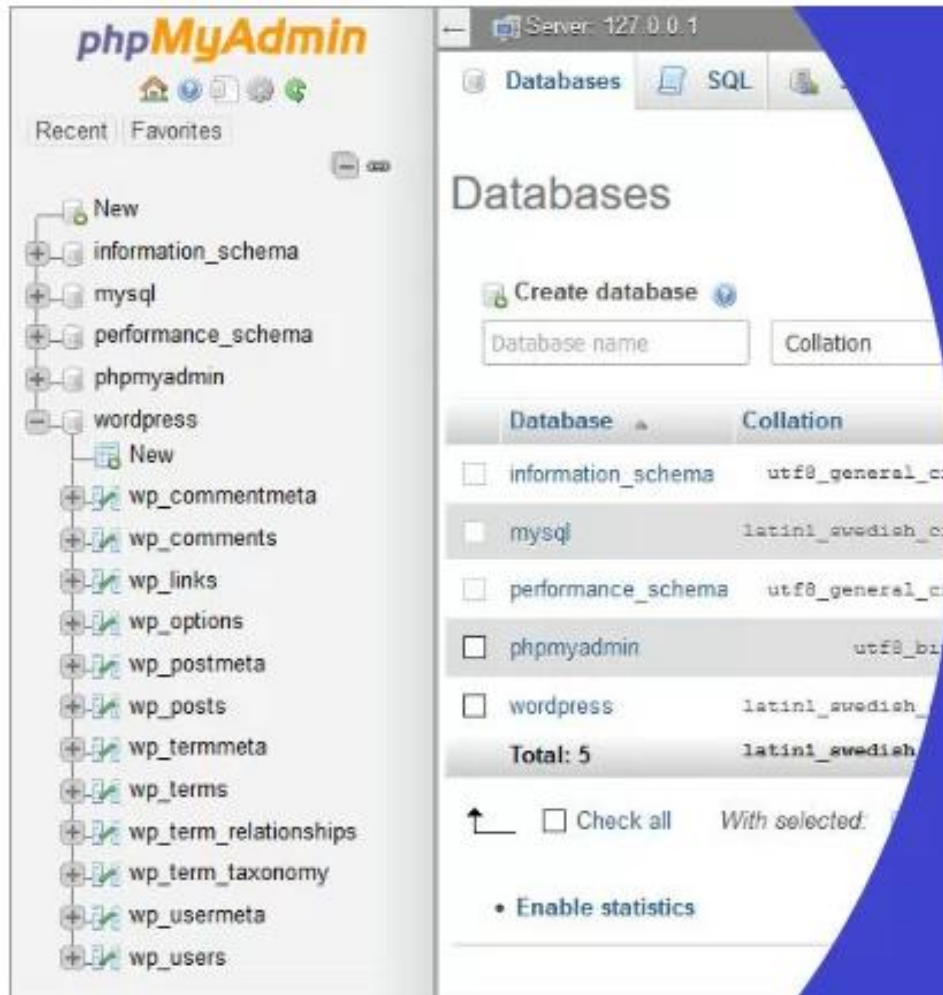
[local-pc]

Bagaimana cara create new database menggunakan CMD dan phpMyAdmin?

A decorative border of green leaves and branches at the top of the page.

1 – Create *Database* in XAMPP using phpMyAdmin

A decorative border of green leaves and branches at the bottom of the page.



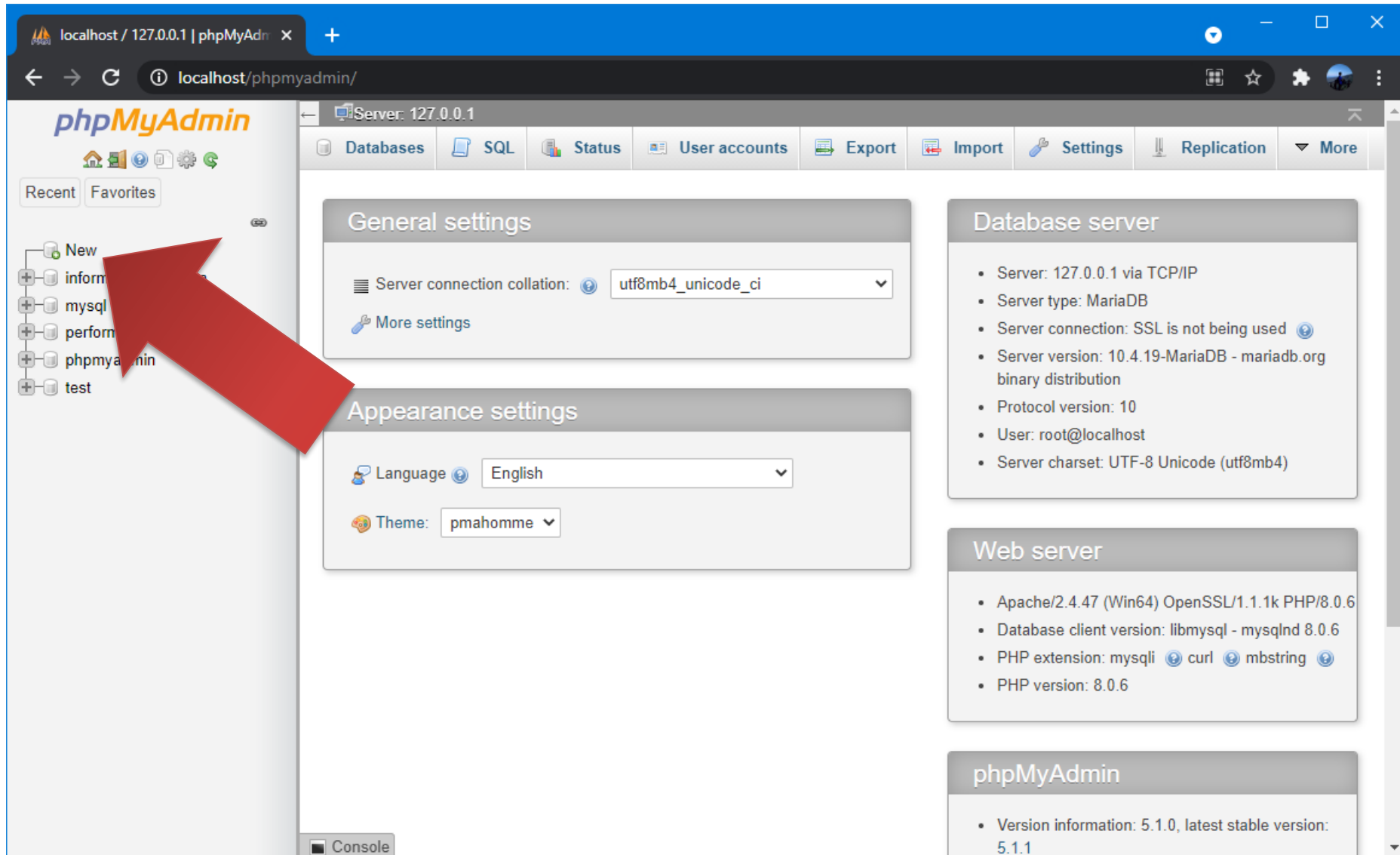
XAMPP phpMyAdmin

How to Create Database and Tables

Recommended to install XAMPP before we start to this lecture. We might need to test our AIMVC project can run locally. If you find any problems related to installing and configuring XAMPP phpMyAdmin, then please find previous lecture in this class or any tutorials to solve it.

Navigate to Database on phpMyAdmin

Navigate cursor to <http://localhost/phpmyadmin/> and click button “new”.



The screenshot displays the phpMyAdmin web interface in a browser window. The address bar shows the URL `localhost/phpmyadmin/`. The interface includes a top navigation bar with tabs for Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, and More. A left sidebar contains a tree view of databases, with the 'New' button highlighted by a large red arrow. The main content area is divided into several panels: General settings (Server connection collation: utf8mb4_unicode_ci), Appearance settings (Language: English, Theme: pmahomme), Database server (Server: 127.0.0.1 via TCP/IP, Server type: MariaDB, Server connection: SSL is not being used, Server version: 10.4.19-MariaDB - mariadb.org binary distribution, Protocol version: 10, User: root@localhost, Server charset: UTF-8 Unicode (utf8mb4)), Web server (Apache/2.4.47 (Win64) OpenSSL/1.1.1k PHP/8.0.6, Database client version: libmysql - mysqlnd 8.0.6, PHP extension: mysqli, curl, mbstring, PHP version: 8.0.6), and phpMyAdmin (Version information: 5.1.0, latest stable version: 5.1.1).

localhost / 127.0.0.1 | phpMyAdmin

localhost/phpmyadmin/

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings Replication More

phpMyAdmin

Recent Favorites

New

inform

mysql

perform

phpmyadmin

test

General settings

Server connection collation: utf8mb4_unicode_ci

More settings

Appearance settings

Language English

Theme: pmahomme

Database server

- Server: 127.0.0.1 via TCP/IP
- Server type: MariaDB
- Server connection: SSL is not being used
- Server version: 10.4.19-MariaDB - mariadb.org binary distribution
- Protocol version: 10
- User: root@localhost
- Server charset: UTF-8 Unicode (utf8mb4)

Web server

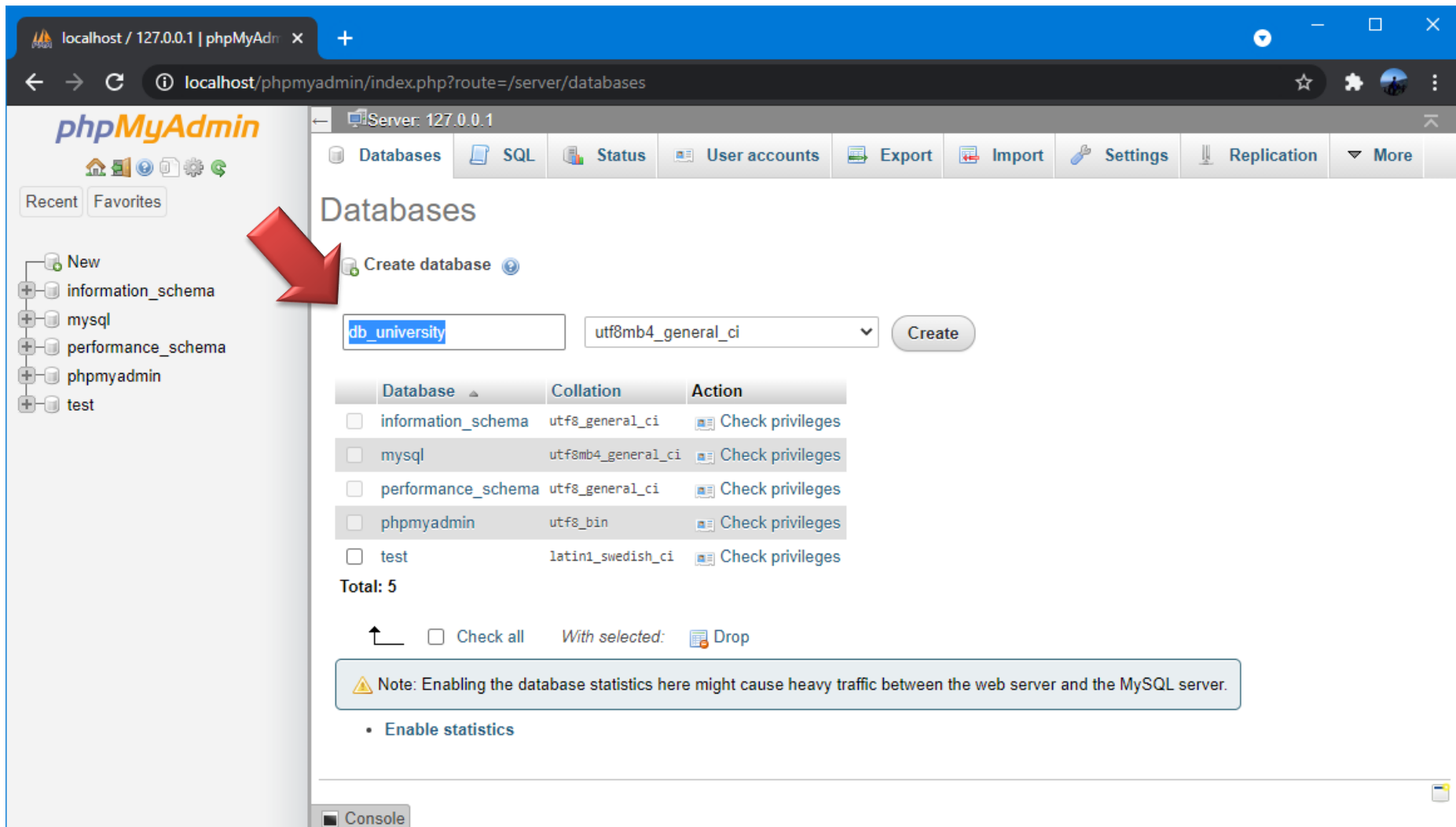
- Apache/2.4.47 (Win64) OpenSSL/1.1.1k PHP/8.0.6
- Database client version: libmysql - mysqlnd 8.0.6
- PHP extension: mysqli curl mbstring
- PHP version: 8.0.6

phpMyAdmin

- Version information: 5.1.0, latest stable version: 5.1.1

Console

Create a Database



The screenshot shows the phpMyAdmin interface for a MySQL server at localhost/127.0.0.1. The 'Databases' tab is selected. A red arrow points to the 'Create database' link. The form for creating a new database is visible, with the input field containing 'db_university' and the collation dropdown set to 'utf8mb4_general_ci'. A 'Create' button is next to the form. Below the form is a table listing existing databases.

Database	Collation	Action
☐ information_schema	utf8_general_ci	Check privileges
☐ mysql	utf8mb4_general_ci	Check privileges
☐ performance_schema	utf8_general_ci	Check privileges
☐ phpmyadmin	utf8_bin	Check privileges
☐ test	latin1_swedish_ci	Check privileges

Total: 5

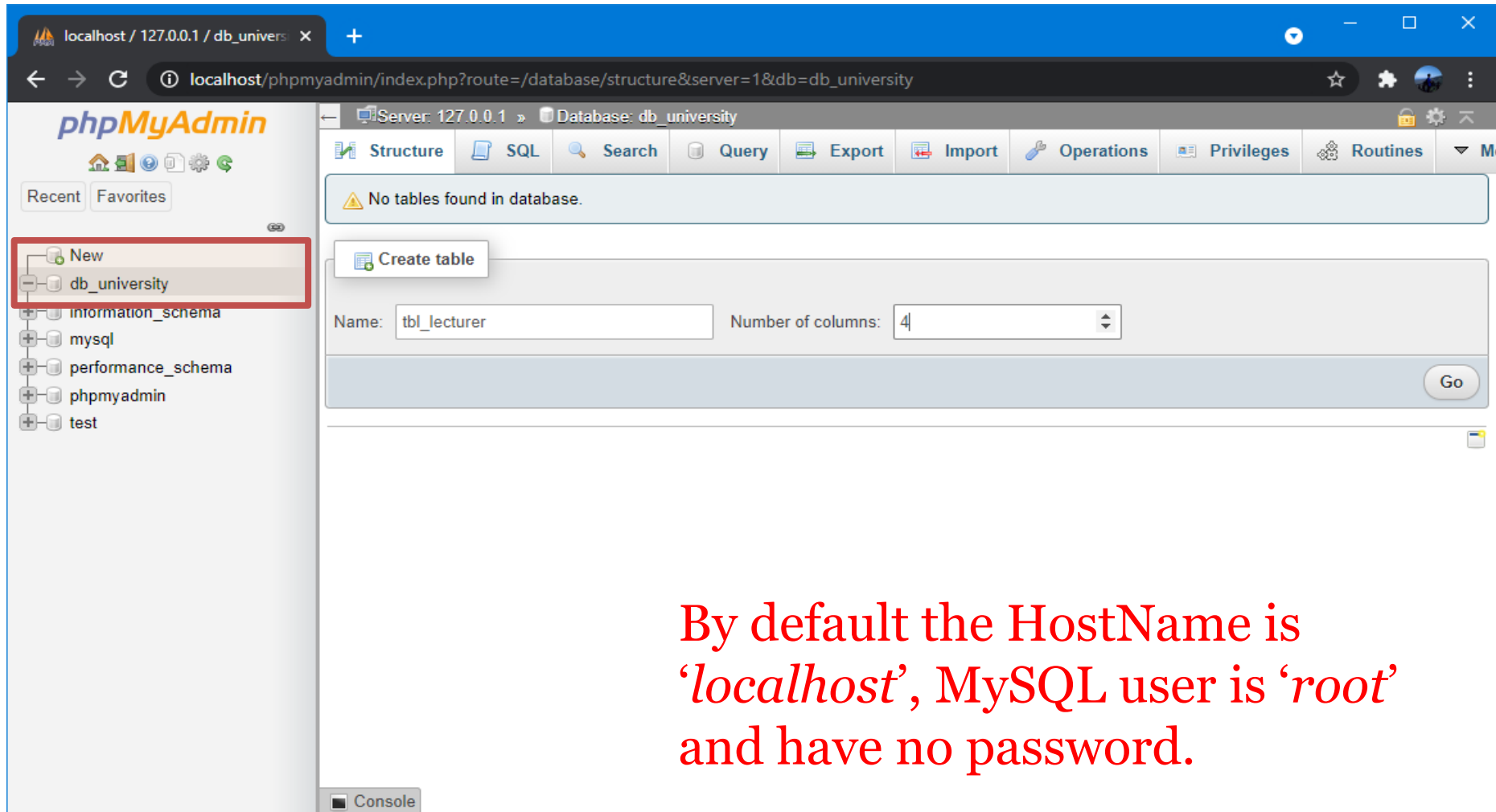
☐ Check all With selected: [Drop](#)

Note: Enabling the database statistics here might cause heavy traffic between the web server and the MySQL server.

- [Enable statistics](#)

See the option to “*Create a Database*” and input field to enter the database name. Write the database name and click the “*Create*” button.

Create Tables Using phpMyAdmin



By default the HostName is 'localhost', MySQL user is 'root' and have no password.

In the phpMyAdmin click on the “Structure” tab and click on the “Create new table on database” wizard to get started. You can see above fields, you can add the **table name** and the total number of fields (*columns*) and click the “Go” button.

Specify Data Type

The screenshot shows the phpMyAdmin interface for a database named 'db_university'. The table 'tbl_lecturer' is selected, and the 'Structure' tab is active. The columns are defined as follows:

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
id	VARCHAR	100	None			☐	---
name	VARCHAR	100	None			☐	---
email	VARCHAR	100	None			☐	---
phone	VARCHAR	100	None			☐	---

Below the column list, the 'Structure' section includes fields for 'Table comments:', 'Collation:', 'Storage Engine:' (set to InnoDB), and 'Connection:'. There is also a 'PARTITION definition:' section with a 'Partition by:' dropdown and a text input for '(Expression or column list)', and a 'Partitions:' text input. At the bottom, there are 'Preview SQL' and 'Save' buttons, and a 'Console' tab is visible.

After the data type, you can specify other optional attributes for each *column*.

Table Structure

The screenshot shows the phpMyAdmin interface for a MySQL database named 'db_university'. The 'Table structure' tab is selected for the table 'tbl_lecturer'. The table has four columns: 'id', 'name', 'email', and 'phone', all of type 'varchar(100)' and collation 'utf8mb4_general_ci'. Below the table structure, there are options to 'Check all', 'With selected', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Fulltext'. There are also buttons for 'Add to central columns' and 'Remove from central columns'. At the bottom, there are sections for 'Indexes' and 'Partitions', both showing 'No index defined!' and 'No partitioning defined!' respectively. The 'Console' tab is visible at the bottom left.

localhost / 127.0.0.1 / db_univers x +

localhost/phpmyadmin/index.php?route=/table/structure&db=db_university&table=tbl_lecturer

Server: 127.0.0.1 » Database: db_university » Table: tbl_lecturer

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Table structure Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	2	name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	3	email	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	4	phone	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#)

[Add to central columns](#) [Remove from central columns](#)

[Print](#) [Propose table structure](#) [Track table](#) [Move columns](#) [Normalize](#)

[Add](#) 1 column(s) after phone [Go](#)

Indexes ⓘ

⚠ No index defined!

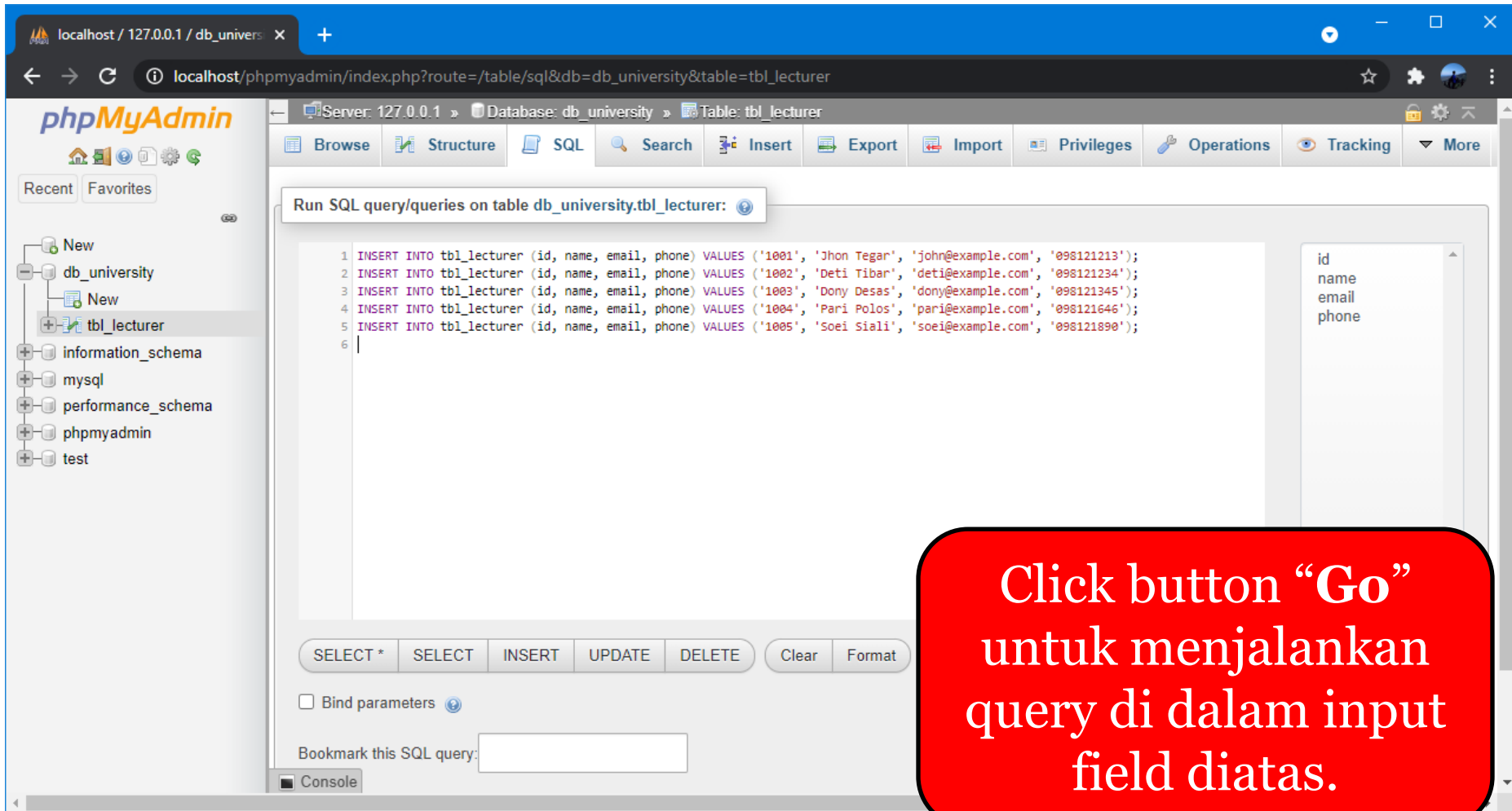
Create an index on 1 columns [Go](#)

Partitions ⓘ

⚠ No partitioning defined!

Console

Insert Data *via* Tab SQL Query



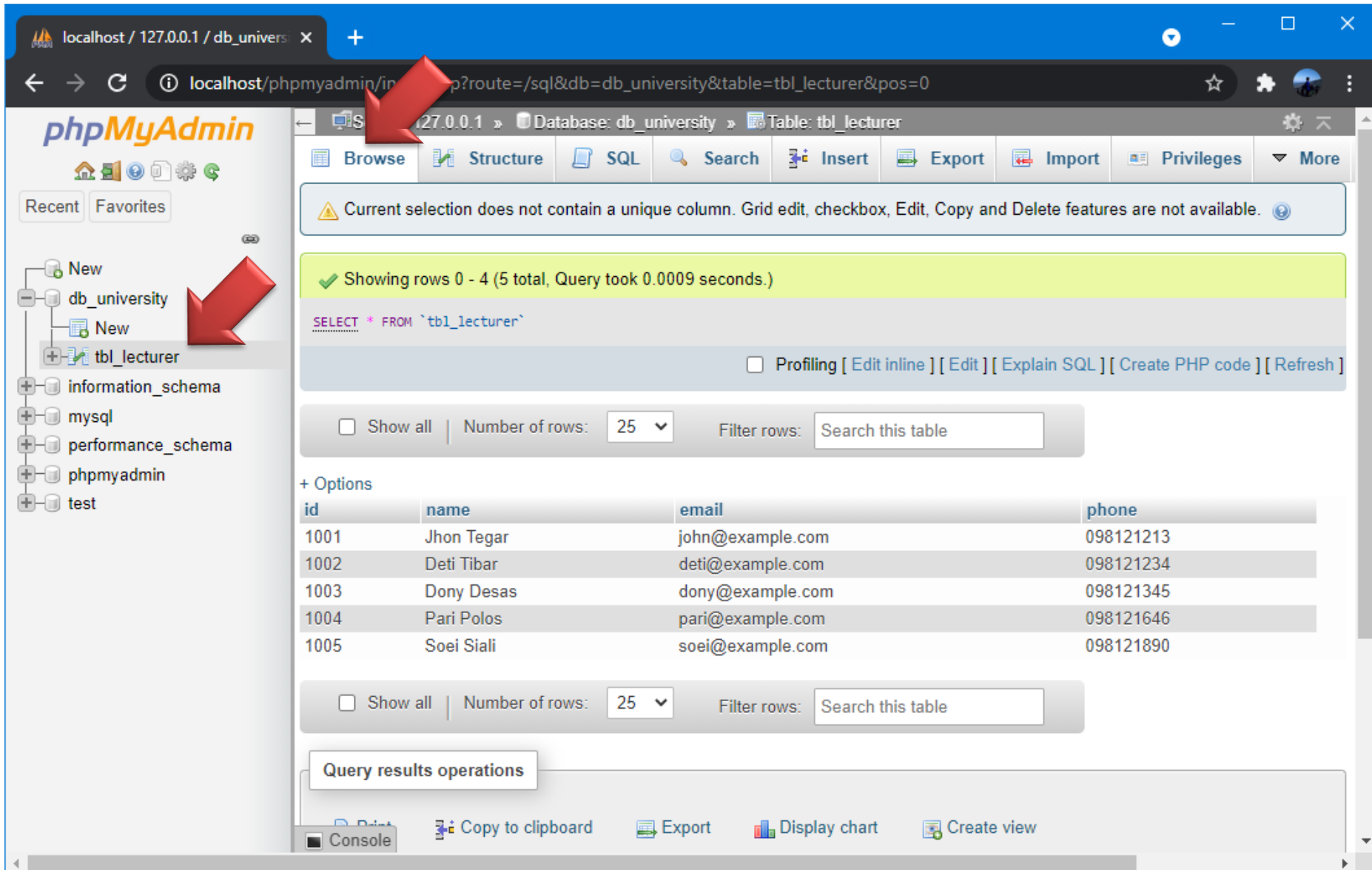
The screenshot shows the phpMyAdmin interface for a database named 'db_university'. The 'tbl_lecturer' table is selected. The SQL tab is active, displaying a query to insert five rows of data. The query is as follows:

```
1 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1001', 'Jhon Tegar', 'john@example.com', '098121213');
2 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1002', 'Deti Tibar', 'deti@example.com', '098121234');
3 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1003', 'Dony Desas', 'dony@example.com', '098121345');
4 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1004', 'Pari Polos', 'pari@example.com', '098121646');
5 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1005', 'Soei Siali', 'soei@example.com', '098121890');
6
```

Below the query, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', and 'Format'. The 'INSERT' button is highlighted. A red callout box with white text says: "Click button 'Go' untuk menjalankan query di dalam input field diatas."

```
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1001', 'Jhon Tegar', 'john@example.com', '098121213');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1002', 'Deti Tibar', 'deti@example.com', '098121234');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1003', 'Dony Desas', 'dony@example.com', '098121345');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1004', 'Pari Polos', 'pari@example.com', '098121646');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1005', 'Soei Siali', 'soei@example.com', '098121890');
```

Data from a MySQL Database



The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL is localhost/phpmyadmin/index.php?route=/sql&db=db_university&table=tbl_lecturer&pos=0. The interface has a sidebar on the left with a tree view of databases: db_university, information_schema, mysql, performance_schema, phpmyadmin, and test. A red arrow points to the 'tbl_lecturer' table under 'db_university'. The main panel shows the 'Table: tbl_lecturer' view. A message at the top states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green bar indicates 'Showing rows 0 - 4 (5 total, Query took 0.0009 seconds.)'. The SQL query 'SELECT * FROM `tbl\_lecturer`' is shown. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A filter section shows 'Show all' and 'Number of rows: 25'. The table data is displayed with columns: id, name, email, and phone. The data rows are: 1001 Jhon Tegar john@example.com 098121213, 1002 Deti Tibar deti@example.com 098121234, 1003 Dony Desas dony@example.com 098121345, 1004 Pari Polos pari@example.com 098121646, and 1005 Soei Siali soei@example.com 098121890. At the bottom, there are links for 'Query results operations', 'Console', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

localhost / 127.0.0.1 / db_univers x +

localhost/phpmyadmin/index.php?route=/sql&db=db_university&table=tbl_lecturer&pos=0

phpMyAdmin

Recent Favorites

db_university

tbl_lecturer

information_schema

mysql

performance_schema

phpmyadmin

test

Database: db_university » Table: tbl_lecturer

Browse Structure SQL Search Insert Export Import Privileges More

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Showing rows 0 - 4 (5 total, Query took 0.0009 seconds.)

SELECT * FROM `tbl\_lecturer`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 Filter rows: Search this table

+ Options

id	name	email	phone
1001	Jhon Tegar	john@example.com	098121213
1002	Deti Tibar	deti@example.com	098121234
1003	Dony Desas	dony@example.com	098121345
1004	Pari Polos	pari@example.com	098121646
1005	Soei Siali	soei@example.com	098121890

☐ Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

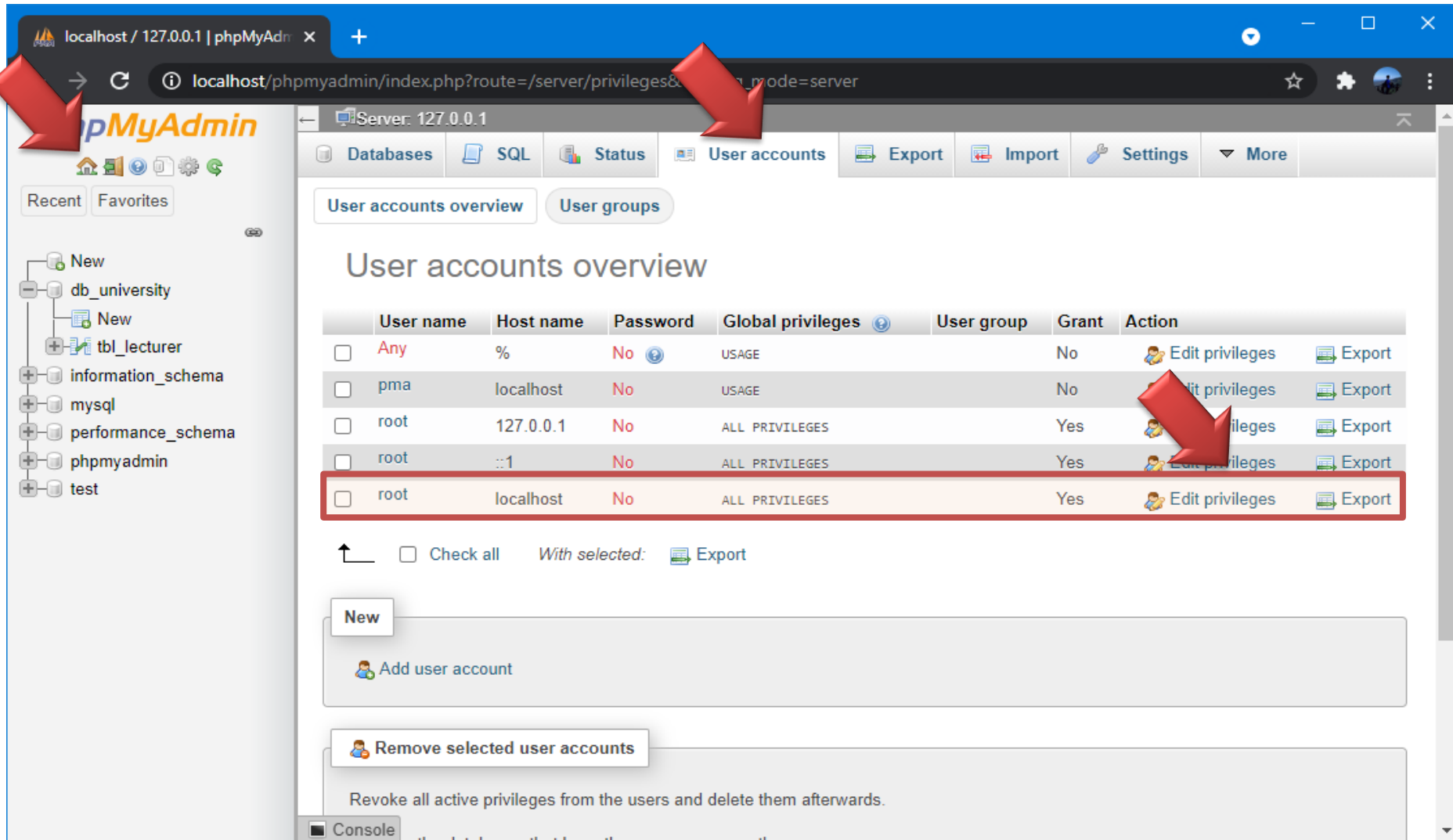
Console Copy to clipboard Export Display chart Create view

A decorative border of green leaves and branches at the top of the page.

2 – Set Password to phpMyAdmin *on* XAMPP

A decorative border of green leaves and branches at the bottom of the page.

User Accounts



The screenshot shows the phpMyAdmin interface for managing user accounts. A red arrow points to the 'User accounts' tab in the top navigation bar. Another red arrow points to the 'Edit privileges' link in the 'Action' column for the 'root' user on 'localhost'. A third red arrow points to the 'root' user on 'localhost' row, which is highlighted with a red border. The interface includes a left sidebar with a database tree, a top navigation bar with tabs like 'Databases', 'SQL', 'Status', 'User accounts', 'Export', 'Import', 'Settings', and 'More'. The main content area shows a table of user accounts with columns: User name, Host name, Password, Global privileges, User group, Grant, and Action.

	User name	Host name	Password	Global privileges	User group	Grant	Action
☐	Any	%	No	USAGE		No	Edit privileges Export
☐	pma	localhost	No	USAGE		No	Edit privileges Export
☐	root	127.0.0.1	No	ALL PRIVILEGES		Yes	Edit privileges Export
☐	root	:::1	No	ALL PRIVILEGES		Yes	Edit privileges Export
☐	root	localhost	No	ALL PRIVILEGES		Yes	Edit privileges Export

Below the table, there are options to 'Check all' and 'With selected: Export'. At the bottom, there is a 'New' section with 'Add user account' and a 'Remove selected user accounts' section with a warning: 'Revoke all active privileges from the users and delete them afterwards.'

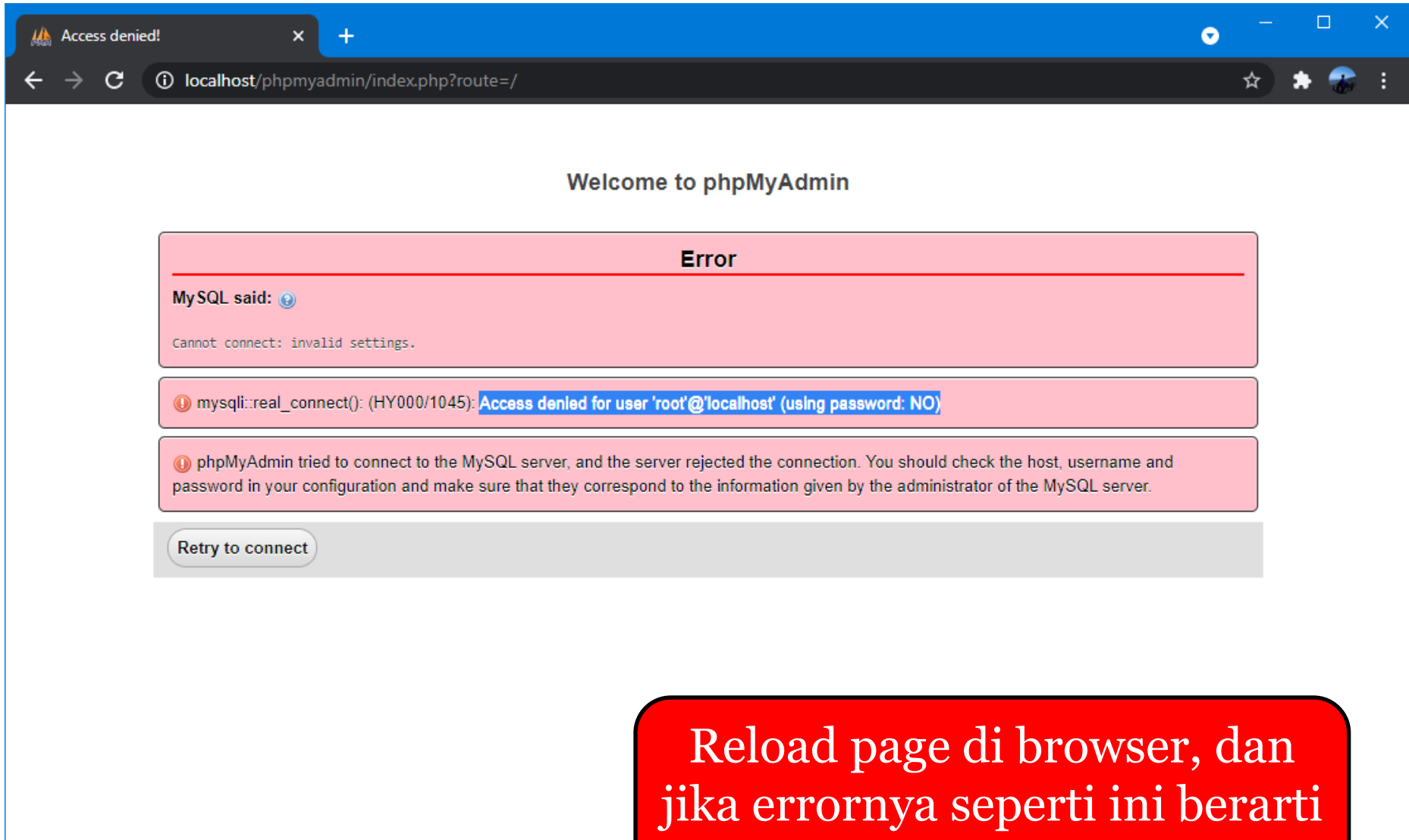
Click on the “user accounts” option at the top of the page and press the “*edit privileges*” under “actions” option for the Username “*root*” and Hostname “*localhost*”.

Change/Edit Password

The screenshot shows the phpMyAdmin web interface in a browser window. The address bar indicates the URL is `localhost/phpmyadmin/index.php?route=/server/privileges&me=root&hostname=localhost&dbname=&tablename=&...`. The interface has a sidebar on the left with a tree view of databases and schemas, including `db_university`, `information_schema`, `mysql`, `performance_schema`, `phpmyadmin`, and `test`. The main content area has a top navigation bar with tabs: `Databases`, `SQL`, `Status`, `User accounts`, `Export`, `Import`, `Settings`, `Replication`, and `More`. Below this, there are sub-tabs: `Global`, `Database`, `Change password` (which is selected), and `Login Information`. The main heading is `Edit privileges: User account 'root'@'localhost'`. A yellow warning box states: `Note: You are attempting to edit privileges of the user with which you are currently logged in.` The `Change password` section contains a radio button for `No Password` and a selected radio button for `Password:`. Below the `Password:` option are two input fields: `Enter:` and `Re-type:`, both containing masked text (dots). To the right of the `Enter:` field is a `Strength:` indicator showing a progress bar and the text `Extremely weak`. Below these fields is a `Password Hashing:` dropdown menu set to `Native MySQL authentication`. At the bottom of the form is a `Generate password` button and a `Generate` button next to an empty input field. A `Go` button is located at the bottom right of the form area. A `Console` tab is visible at the bottom left of the interface.

Choose the third tab “*change password*” and type your password in the provided field, retype the password to confirm it and then finally click on the “Go” button to conclude the process.

Error Access Denied

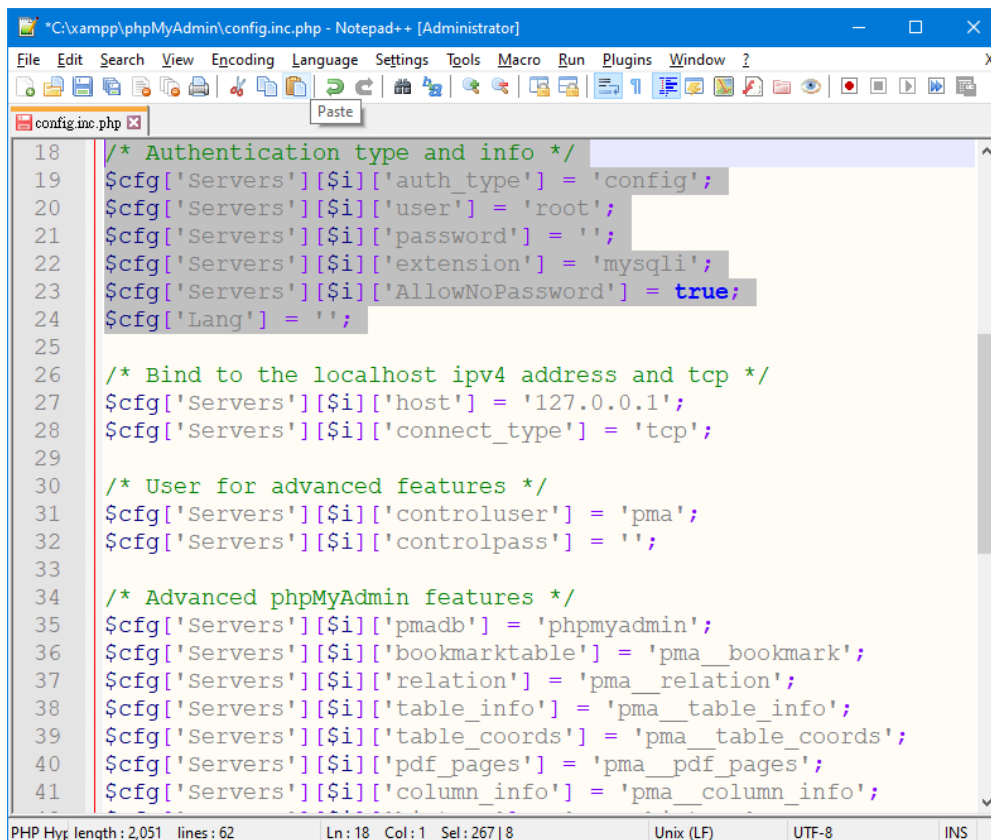


Reload page di browser, dan jika errornya seperti ini berarti step awal sudah benar.

Configuration Settings [*allow without password*]

ERROR #1045 - Access denied for user 'root'@'localhost' (using password: NO) when opening http://localhost/phpmyadmin.

- ❖ Navigate to "C:\xampp\phpMyAdmin"
- ❖ Find the file "config.inc.php" and open in a Notepad or better, Notepad++ (**run as administrator**).
- ❖ At line # 18 to 24, is the settings for user authentication.



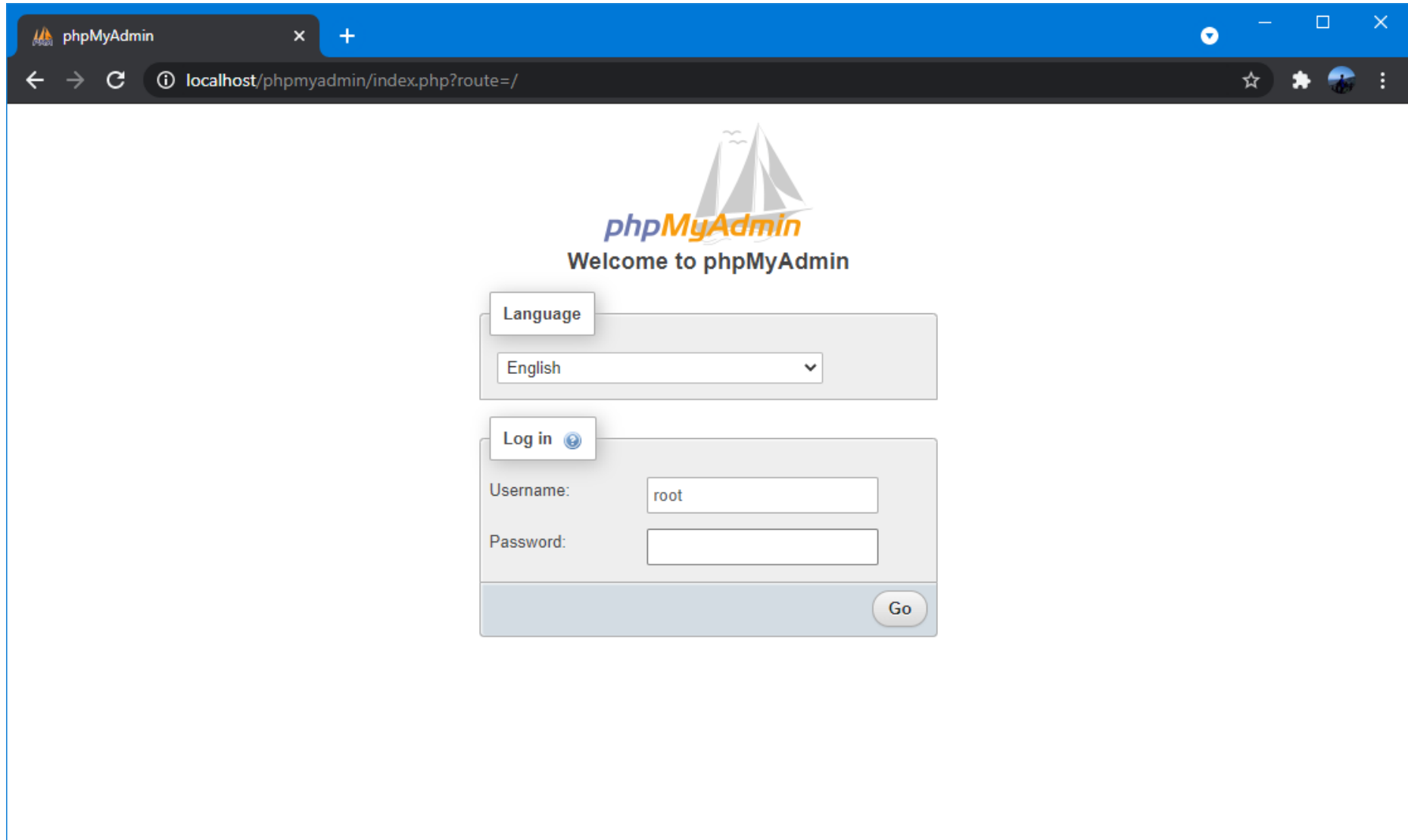
```
18  /* Authentication type and info */
19  $cfg['Servers'][$i]['auth_type'] = 'config';
20  $cfg['Servers'][$i]['user'] = 'root';
21  $cfg['Servers'][$i]['password'] = '';
22  $cfg['Servers'][$i]['extension'] = 'mysqli';
23  $cfg['Servers'][$i]['AllowNoPassword'] = true;
24  $cfg['Lang'] = '';
25
26  /* Bind to the localhost ipv4 address and tcp */
27  $cfg['Servers'][$i]['host'] = '127.0.0.1';
28  $cfg['Servers'][$i]['connect_type'] = 'tcp';
29
30  /* User for advanced features */
31  $cfg['Servers'][$i]['controluser'] = 'pma';
32  $cfg['Servers'][$i]['controlpass'] = '';
33
34  /* Advanced phpMyAdmin features */
35  $cfg['Servers'][$i]['pma_db'] = 'phpmyadmin';
36  $cfg['Servers'][$i]['bookmarktable'] = 'pma_bookmark';
37  $cfg['Servers'][$i]['relation'] = 'pma_relation';
38  $cfg['Servers'][$i]['table_info'] = 'pma_table_info';
39  $cfg['Servers'][$i]['table_coords'] = 'pma_table_coords';
40  $cfg['Servers'][$i]['pdf_pages'] = 'pma_pdf_pages';
41  $cfg['Servers'][$i]['column_info'] = 'pma_column_info';
```

Change the values to :

```
/* Authentication type and info */
$cfg['Servers'][$i]['auth_type'] = 'cookie';
$cfg['Servers'][$i]['user'] = 'root';
$cfg['Servers'][$i]['password'] = '12345';
$cfg['Servers'][$i]['extension'] = 'mysqli';
$cfg['Servers'][$i]['AllowNoPassword'] = false;
$cfg['Lang'] = '';
```

password: password is not set in the configuration file.

Testing



The image shows a web browser window with the phpMyAdmin interface. The browser's address bar displays the URL `localhost/phpmyadmin/index.php?route=/`. The phpMyAdmin logo, featuring a sailboat, is centered at the top of the page. Below the logo, the text "Welcome to phpMyAdmin" is displayed. The interface includes a "Language" dropdown menu currently set to "English". Below this is a "Log in" button with a help icon. The login form contains two input fields: "Username:" with the value "root" and "Password:" which is empty. A "Go" button is located at the bottom right of the login form.

phpMyAdmin

localhost/phpmyadmin/index.php?route=/

phpMyAdmin

Welcome to phpMyAdmin

Language

English

Log in

Username: root

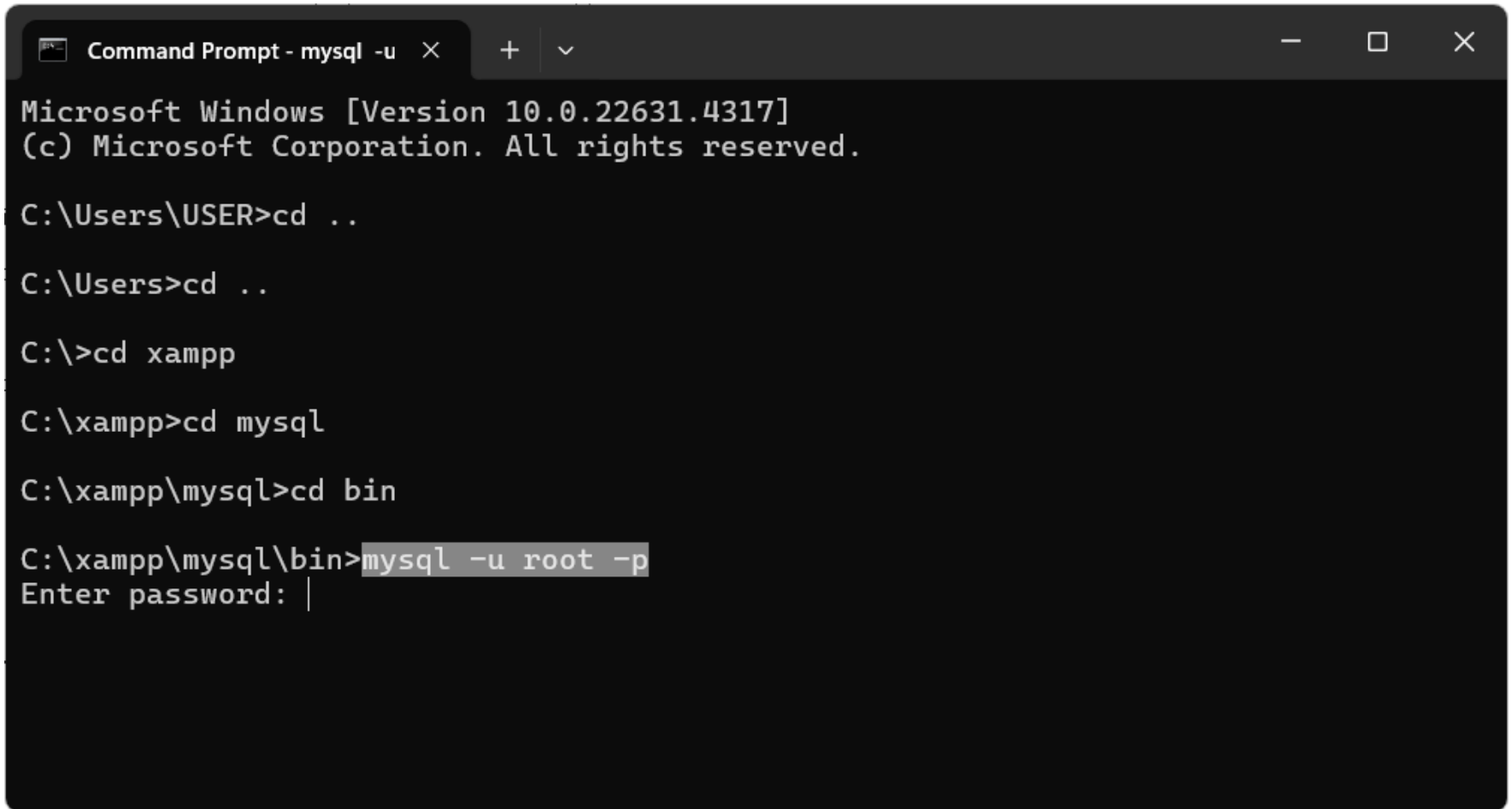
Password:

Go

Bagaimana cara login ke
database dengan
meggunakan CMD?

Pada tahap awal silahkan masuk ke MySQL :

- Masuk ke folder “C:/xampp/mysql/bin”
- Buka CMD dan jalan command:
>mysql -u root -p



```
Command Prompt - mysql -u  X + v
Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd ..

C:\Users>cd ..

C:\>cd xampp

C:\xampp>cd mysql

C:\xampp\mysql>cd bin

C:\xampp\mysql\bin>mysql -u root -p
Enter password: |
```

Anda mencoba masuk ke database dengan menggunakan user “root” dan password.

Pada tahap awal silahkan masuk ke MySQL :

- Masuk ke folder “C:/xampp/mysql/bin”
- Buka CMD dan jalan “mysql -u root -p”

```
mysql> CREATE DATABASE db_unklab;  
mysql> CREATE DATABASE db_unklab;  
Query OK, 1 row affected (0.78 sec)  
  
mysql> |
```

mysql> SHOW DATABASES;

```
mysql> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| db_attendance |  
| db_unklab |  
| information_schema |  
| mysql |  
| performance_schema |  
| sys |  
| testadmin |  
+-----+  
7 rows in set (0.00 sec)
```

mysql> use db_unklab;

```
mysql> use db_unklab;  
Database changed  
mysql> |
```

Show struktur table:

mysql> DESCRIBE tbl_customer;

```
CREATE TABLE nama_tabel (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    nama VARCHAR(50),  
    umur INT,  
    alamat VARCHAR(255)  
);
```

SETELAH ITU BUAT TABEL DAN ISI DATA PADA TABEL YANG TELAH DIBUAT.

mysql> CREATE TABLE tbl_students (reg_number VARCHAR(255), nim_number VARCHAR(255), email VARCHAR(255), fullname VARCHAR(255), password VARCHAR(255), created_at VARCHAR(255), updated_at VARCHAR(255));

mysql> INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at) VALUES

-> ('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),

-> ('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),

-> ('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),

-> ('1004', '1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', NOW(), NOW()),

-> ('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());

```
mysql> CREATE TABLE tbl_students (reg_number VARCHAR(255), nim_number VARCHAR(255), email VARCHAR(255), fullname VARCHAR(255), password VARCHAR(255), created_at VARCHAR(255), updated_at VARCHAR(255));  
Query OK, 0 rows affected (2.46 sec)
```

mysql> INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at) VALUES

-> ('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),
-> ('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),
-> ('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),
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-> ('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());

```
Query OK, 5 rows affected (0.28 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> |
```

Insert satu baris data ke dalam tabel

```
INSERT INTO nama_tabel (nama_kolom1, nama_kolom2, nama_kolom3)  
VALUES ('nilai1', 'nilai2', 'nilai3');
```

Contoh insert table

```
INSERT INTO tbl_mahasiswa (nim, nama, jurusan) VALUES ('123456', 'John  
Doe', 'Teknik Informatika');
```

```
MySQL 8.0 Command Line Client  
mysql> CREATE TABLE Student (  
->   Stud_ID int AUTO_INCREMENT PRIMARY KEY,  
->   Name varchar(45) DEFAULT NULL,  
->   Email varchar(45) NOT NULL UNIQUE,  
->   City varchar(25) DEFAULT NULL  
-> );  
Query OK, 0 rows affected (0.88 sec)  
  
mysql> INSERT INTO Student(Stud_ID, Name, Email, City)  
-> VALUES (1,'Stephen', 'stephen@javatpoint.com', 'Texax'),  
-> (2, 'Joseph', 'Joseph@javatpoint.com', 'Alaska'),  
-> (3, 'Peter', 'Peter@javatpoint.com', 'california');  
Query OK, 3 rows affected (0.18 sec)  
Records: 3  Duplicates: 0  Warnings: 0  
  
mysql> SELECT * FROM Student;  
+-----+-----+-----+-----+  
| Stud_ID | Name   | Email                               | City       |  
+-----+-----+-----+-----+  
|      1 | Stephen | stephen@javatpoint.com | Texax     |  
|      2 | Joseph | Joseph@javatpoint.com  | Alaska    |  
|      3 | Peter  | Peter@javatpoint.com   | california |  
+-----+-----+-----+-----+
```

[server]

Bagaimana cara create new
database di server side
menggunakan Putty?

Password for User Root

Disini ada beberapa tahap yang harus kita lakukan:

- `sudo su` (masuk ke server via putty atau cmd)
- **`sudo mysql`**
..(jika sudah masukan password, step di bawah di skip saja)
- **`mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'ganti_password_kita';`**
- **`mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH auth_socket;`**

Kemudian keluar dari MySQL prompt
`mysql> exit`

Seharusnya, User Root sekarang ini harus required password untuk akses database:

- **`mysql -u root -p`**
... (new your password)

Pada tahap awal silahkan masuk ke MySQL :

- Buka CMD, masuk ke server dan jalan command:
 - > sudo su
 - > mysql -u root -p

```
david@websiteapache:~$ mysql david -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 124
Server version: 5.7.20-0ubuntu0.17.10.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database users;
Query OK, 1 row affected (0.00 sec)

mysql> show databases;
+-----+
| Database                |
+-----+
| information_schema      |
| david                   |
| mysql                   |
| performance_schema      |
| phpmyadmin              |
| sys                     |
| test_database           |
| users                    |
+-----+
8 rows in set (0.01 sec)

mysql> quit
Bye
```

Jika belum mempunyai user mysql yang baru, anda akan mencoba masuk ke database dengan menggunakan default user “*root*” dan password.

Pada tahap awal silahkan masuk ke MySQL :

- Masuk ke folder “C:/xampp/mysql/bin”
- Buka CMD dan jalan “mysql -u root -p”

```
mysql> CREATE DATABASE db_unklab;  
mysql> CREATE DATABASE db_unklab;  
Query OK, 1 row affected (0.78 sec)  
  
mysql> |
```

mysql> SHOW DATABASES;

```
mysql> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| db_attendance |  
| db_unklab |  
| information_schema |  
| mysql |  
| performance_schema |  
| sys |  
| testadmin |  
+-----+  
7 rows in set (0.00 sec)
```

mysql> use db_unklab;

```
mysql> use db_unklab;  
Database changed  
mysql> |
```

Show struktur table:

mysql> DESCRIBE tbl_customer;

```
CREATE TABLE nama_tabel (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    nama VARCHAR(50),  
    umur INT,  
    alamat VARCHAR(255)  
);
```

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| Stud_ID | Name   | Email                               | City      |  
+-----+-----+-----+-----+  
|      1 | Stephen | stephen@javatpoint.com | Texax    |  
|      2 | Joseph  | Joseph@javatpoint.com | Alaska   |  
|      3 | Peter   | Peter@javatpoint.com   | california |  
+-----+-----+-----+-----+
```

Langkah-langkah untuk mengakses dan mengelola database MySQL via command line (CMD/Terminal) di Ubuntu 22.04 Server, mulai dari membuka MySQL, membuat database dan tabel, hingga menyisipkan data:

1. Login ke MySQL:

```
> sudo su  
> sudo mysql -u root -p
```

Masukkan password MySQL root saat diminta.

2. Tampilkan Semua Database:

```
> SHOW DATABASES;
```

3. Create New Database:

Misalnya, buat database bernama school_db:

```
> CREATE DATABASE school_db;
```

Lalu gunakan database tersebut:

```
> USE school_db;
```

4. Create New Tabel tbl_student:

```
CREATE TABLE tbl_student (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  age INT,  
  grade VARCHAR(10)  
);
```

5. Tampilkan Struktur Tabel:

```
DESCRIBE tbl_student;
```

6. Insert Data ke Tabel

```
INSERT INTO tbl_student (name, age, grade)  
VALUES  
( 'Andi', 16, '10A'),  
( 'Budi', 17, '11B'),  
( 'Citra', 15, '9C');
```

7. View Data di Tabel

```
SELECT * FROM tbl_student;
```

**Buat
database (db_unklab)
dan table (tbl_mahasiswa)
yang sama di Local PC dan
di Ubuntu server.**

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

